

Lab Tasks

MI102 - Network Systems and Security

Fachhochschule Kiel

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Task 6 (1)

- Install a DBMS
- Install Keycloak
 - Use the DBMS to store persistent data.
 - Hints:
 - Keycloak and the DBMS can be run using Docker
 - *Production Mode* is not necessary
 - SSL is not necessary when you work with local network
 - use *Developer Mode*

Task 6 (2)

- Develop a concept for an application that stores personal data on a server.
 - It shall be a multi-user application.
 - Therefore, each user shall log in prior using the application.
 - Each user shall be able to read and to store only his own data on the server.

Task 6 (3)

- OAuth2 offers many authentication flows.
- You shall use *Authorization Code Flow*.

Task 6 (4)

- Examples for applications:
 - ToDo-Lists
 - Personal Notebook
 - Guest List
 - Phone Book
 - Calendar

Task 6 (5)

- Develop a web client for your application.
 - The web client shall allow to read and enter personal data of one user via a web page.
 - Authorization shall be done via Keycloak.
 - Tokens shall be re-used in all pages of your application. That means during page reloads, in new tabs or if you restart your browser.
 - If the tokens expire, revalidate them against the Keycloak server without requesting a new login from the user.
 - Use PKCE!

Task 6 (6)

- Develop an API for your application.
 - The API shall allow to request (GET) and to store (POST) personal data.
 - Authorization shall be done via Keycloak.
 - Ensure that each user only sees his own data.
 - Example:
User *Tim* shall only see personal data of user *Tim*.
 - Validate all tokens

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